Image-to-patient MES evaluation on LIMUC

ViT-B/16 and Swin-B use image-label supervision; evaluation paths remain distinct.

A PATIENT-GROUPED NESTED DEVELOPMENT EVALUATION

Development cohort

9,590 images / 478 patients
5 outer folds; hold out one
Keep each patient together

→

4-fold inner selection

Select loss, aggregation, family and refit duration
Only outer-training patients

→

Refit and predict

Refit outer-training data
Predict held-out outer fold
No outer-guided selection

Pool all five outer folds separately per seed; patient scoring uses modal-low (C).

Six aggregation candidates; all outer folds selected mean probability followed by argmax.

B LOCKED OFFICIAL-PARTITION RE-EVALUATION

Official test images

1,686 images; 86 patients
Previously examined

→

15 checkpoints per family

5 development folds × 3 seeds
Historical fixed ensemble

→

Two views per checkpoint

Original + horizontal flip
Mean of 30 softmax vectors

↓

Image probabilities

Four-class image vector
Separate ViT and Swin probability averages

→

Image argmax

→

Image-level scoring

Compare with image labels

→

Patient mean → argmax

Average image probabilities

→

Patient-level scoring

Compare with modal-low (C)

Official labels enter scoring only during re-evaluation; these are not nested outer refits.

C REFERENCE CONSTRUCTION IS SEPARATE FROM PREDICTION

Released image labels

Expert MES labels 0–3

→

Modal-low reference

Modal grade; lower-grade ties

→

Patient-level comparison

Predictions from A or B

Patient references summarize image bags, not independently adjudicated examination-level MES.